

Principles & Ethics in Research

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Some Good Resources for NNDL

- Elizabeth DePoy, “Introduction to Research”, 7th Edition, Elsevier, 2015.
- C. R. Kothari, “Research methodology”, New Age International Publishers, 2004.
- Rupinder Tewari & Mamta Bhardwaj “Intellectual Property: A Primer for Academia”, Punjab UNiversity, 2021.

Organization

- Introduction
- What is Research?
- What is innovation?
- How to meet these objectives?
- Principles for Research & Innovation
- Ethics in Research & Innovation
- Best practices
- Plagiarism
- True pursuit of research & innovation

Introduction

- Research & innovation looks like two faces of a coin, where former referring to invention
- However, there is a need to understand the differences between them and later use them interchangeably, if required
- Principles and ethics plays a very important in the digital era

What is Research?

- **Philosophical** : Research is the search for universal principles and tries to understand reality.
- **Working**: Research is systematic investigation into and study of materials and sources in order to establish facts and reach new conclusions
- **Working**: Research is creative and systematic work undertaken to increase the stock of knowledge
- **Working**: Research is to observe and understand what everyone else has done so far and do what none have done so far

What is Innovation?

- **Philosophical:** Search for product and processes that can solve a specific problem to provide solutions that are cheaper, faster, and better in a given context.
- **Working:** Makes some small variation to existing product and process to get next version

How to meet these objectives?

- We need to really understand the distinctions between invention (research) and innovation
- After that the strategies to carryout is planned accordingly
- Research seems to be a long term strategy
- Innovation seems to be a short term strategy
- As long as these distinctions are clear, they may be used interchangeably

Principles for Research & Innovation

- Some guiding principles to be drawn to carryout R&I
- R&I is an important contribution to the development of society
- Realizing R&I is different from course based study
- Uncertainty is part of R&I field
- Enjoy the journey rather than end result that is being sought
- Consistent hardwork is of paramount interest than sporadic intelligent spiky activities
- Patience and systematic approach gives end results
- Appreciate each part of the R&I program



Ethics in R&I

- Pursuit of truth and no hurry to get results as hypothesized
- No manipulation of experiments
- No manipulation of theoretical, simulation or experimental results
- No copying of content from existing literature
- Even, if needed, properly acknowledged
- Hypothesis, analyze results, ready accept findings
- Findings themselves may give a near direction for research

Ethics in R&I

- Writing articles independently
- Project existing or earlier work properly. Do not misguide.
- Based on that bring out proper hypothesis
- Consider reviews of articles seriously
- Revise your work sincerely
- No errors either in experiments, presentations and thesis
- Better be late than ethically wrong things that will haunt for life

True Pursuit of Research

- Research as a random or uncertain process
- Course work, Literature review
- Identifying few directions, Carrying out research work
- Documenting and publishing, Addressing reviewer comments
- Making story for a thesis, Synopsis
- Thesis submission, Thesis defence
- Reading articles, exploration, documenting and publishing, defending
- Taking care : Physical, mental, patience and positive thinking
- Guiding research scholars
- Carrying out independent research

Research Methodology

- **Research** : Creative and systematic work undertaken to increase the stock of knowledge.
- **Methodology** : A system of methods used in a particular area of study or activity.
- **Research Methodology** : Creative and systematic work undertaken to increase the stock of knowledge in a particular area of study.
- **Is there any methodology for research ?**
- **Process** : Whether this process is **Deterministic or Random?**
- **Objective** : Want to convince that research is a random process.
- If research is a random process, then methodology makes sense !



Components of Research

- **Core components** : Reading literature, performing Experiments (pen-paper, simulation, workbench), Documenting findings (RED).
- **Research based degree**: UGR, MS(R), MTech(R), MSc(R), PhD etc.
- **Criteria for degree** : Research based thesis.
- **Research based Degree** : Course work + RED + Thesis
- PhD takes about 4 to 5 years is a fact, but varies from one to other !
- Agree all of the above components, not sure how much time each component takes place.

Preparedness for a Research based Degree

- **Different** : Different from a course based degree like BTech / MTech.
- **Well laid out** : Components are well laid out.
- **Components** : Course work + RED + Thesis
- **Commitment** : As a scholar committed to going through each component sincerely.
- **Questioning** : Question the findings in the literature, not the process (components).
- **Success** : Investing time, hard work, positive bent of mind, never give up attitude.
- **Relationship** : Cordial relationship with supervisor and group members.



Research based Degree: **Process**

- Course work: **Getting raw materials to your research**
- Comprehensive exam: **Analytical thinking & generalization**
- State of the art seminar: **Literature review, Critical review & problem identification**
- Annual progress: **Reporting new explorations & publications**
- Synopsis: **Brief report (bird's view) on various explorations.**
- Thesis submission: **Extended report in unified framework to tell story.**
- Each component is important but no exact timeline.



Research based Degree (RBD): How to get maximum benefit?

- Consider each stage in its true spirit
- Set of good courses and performing well
- Concious attempt to pick up fundamentals, ability to answer questions related to them & also generalization
- Literature: review vs critical review
- Doing something new & communicating (written & oral)
- Thesis: How to make a story out of explorations?
- Research: Journey, summary, conclusions & future scope



RBD: How to measure your progress? (Optimistic)

- Courses by the end of first year
- Comprehensive by one & half years
- Problem identification by the end of two years
- At least one new exploration & one paper by the end of three years
- At least two more explorations & two papers by the end of three & half years
- Synopsis by the end of three year nine months
- Thesis submission by the end of four years
- Happens in 10-20% cases

RBD: How to measure your progress?(Most likely)

- Courses by the end of first year
- Comprehensive by one & half years
- Problem identification by the end of two and half years
- At least one new exploration & one paper by the end of three and half years
- At least two more explorations & two papers by the end of four & half years
- Synopsis by the end of four year nine months
- Thesis submission by the end of five years
- Happens in 50-60% cases



RBD: How to measure your progress?(Few cases)

- Courses by the end of one and half years
- Comprehensive by two & half years
- Problem identification by the end of three and half years
- At least one new exploration & one paper by the end of four years
- At least two more explorations & two papers by the end of five years
- Synopsis by the end of five and half years
- Thesis submission by the end of six years
- Happens in 10-20% cases

RBD: What is Expected after RBD?(Pleasure)

- Learnt the art of doing research
- Carried out quality work
- Ability to critically review
- Ability to think differently
- Ability to take risk
- Ability to articulate (spoken) clearly
- Ability to communicate (written) well

RBD: Ensuring Quality

- Too many PhDs
- However good work always gets recognition
- Survival of the fittest
- See what everyone else is doing & do what no body has done
- While entering you are 24 carrot gold bar
- While going out you should become ornaments so that some one will wear you
- Unless gold bar does not take goldsmith hammer beats, it can never become ornament
- Keep this in mind through out your PhD program journey



What is Important for Research?

- Passion to explore new things
- Investment of time
- Working hard
- Discipline to spend time regularly
- Reading good articles
- Thinking over them and generating new ideas
- Development of patience

What is Definitely not Research?

- Writing programs
- Using toolkits and packages
- Relying only on computers and machines
- Simply doing experiment after experiment without proper thought process
- Publishing in low quality conferences and journals
- Roaming from one conference to other
- Spending time on non-issues



Am I competent enough to do RBD?

- Any person with common sense
- Any person who has passion
- Any person who has an urge to excel
- Any person who can invest time
- Any person who is interested in developing himself to better state of mind

Role of Supervisor

- Known vs unknown paths
- RBD process is not travelling using gps and maps
- Scholar plans to make a journey from point A to point B
- He does not have GPS and maps
- Gets lost at many intermediate points
- Asks stranger to move further
- Supervisor plays the role of the STRANGER
- Each scholar is different and each PhD guidance is a new process



What you should avoid and definitely do?

- Avoid developing complaining attitude.
- Avoid comparing with others.
- Definitely help each other among friends.
- Definitely your growth depends on your investment.
- Avoid your interest to follow e^{-t} .
- Definitely try hard for your interest to follow e^t
- At least try to keep your interest to follow $e^{-t} + e^t$

Taking care of yourself

- **Physical** : Unless you are doing labor intensive experimental work, you need physical activity to take care of your body.
- **Mental** : Research needs lot of mental activity. You need rejuvenation activity to take care of your mind.
- **Patience** : Uncertainty is the only deterministic thing in Research! You need to cultivate the skill of developing patience over aberrations.
- **Positive Thinking** : Positive thinking is the backbone for the success in research.

Guiding Research Scholars

- **Intelligence** : More than good academic background, maturity and emotional intelligence are more important for a scholar.
- **Uniqueness** : Each scholar is unique and needs a different way of mentoring.
- **Your Maturity** : You are senior among the two and ensure you have more maturity and understanding.
- **Difference of handling** : Two adults are interacting face to face over 4 to 5 years ! How to ensure good partnership ?
- **Nature of Supervision** : Never do spoon feeding at the same time don't be yourself away from the field.
- **Goldsmith** : Add value to the gold bar (scholar) from your workmanship as a goldsmith.

Carrying out Independent Research

- **Researcher in You** : Lead from front, not research manager, you are a researcher !
- **Independent Researcher** : Read literature, listen to talks and ponder for new ideas. Implement some of them, may be in collaboration.
- **Relevance** : You should keep yourself relevant in the changing research landscape.
- **Don't allow Inquisitiveness to die** : Ability to question what is happening in the field and thinking differently.
- **Don't Age** : You may age physically, but not mentally. Keep the child like ability to question your field.
- **Life long** : You are a life long student and researcher by yourself.



Why Research is a Random Process

- If you recall most points discussed you may agree.
- You may feel the relevance to achieve greater heights in research.
- However, it is difficult to put any time line for each of these points.
- Hence it is a life long process.
- Hence the random process nature of the research field.
- Thus you will avoid confusing research methodology with a deterministic process.

Summary

- Research is a mental activity that keeps you interesting to explore.
- Research is a random process but its broad components can be studied via research methodology.
- All different components are well laid out through a careful thought process.
- Invest time and effort with utmost sincerity and best effort to excel.
- Consistent hardwork pays more compared to intelligence.
- Whether industry, academia or life at large, research is a must.

Thank You